CY TECH

Team #28

ASU Interplanetary Initiative
Arizona State University

SpaceHACK for Sustainability

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Track: Environmental Migration in Sahelian Africa

Our Approach & Data Analysis

Step 1: Understanding the Context

- After following the trainings, we conducted research on the Sahelian Africa crisis.
- We identified key factors related to water scarcity and its consequences.

Step 2: Key Focus Areas

🔍 Evolution over time – How has the water crisis worsened? Focus on the period from 2000 to 2020.

💧 Water availability – Tracking fluctuations in water resources.

🌍 Drought & Land Impact – Understanding desertification and soil degradation.

Step 3: Data & Sources

📊 Our datasets:

- *CIESIN*
- *NASA's DAAC LP*
- *JRC*
- *ERA5*
- *GPM*

Other Sources : CNES (Centre national d'études spatiales), AMMA-CATCH (Observatoire éco-hydro-météo en Afrique de l'Ouest), Le Monde…

Initial Results & Insights

🔍 **Key Findings**

- Sahelian Africa is a semi-arid region with a short wet season and an extended dry season.
- Preliminary drought analysis shows a decline in vegetation in the Sahel zone since 2000.

🚀 **Opportunities for Further Exploration**

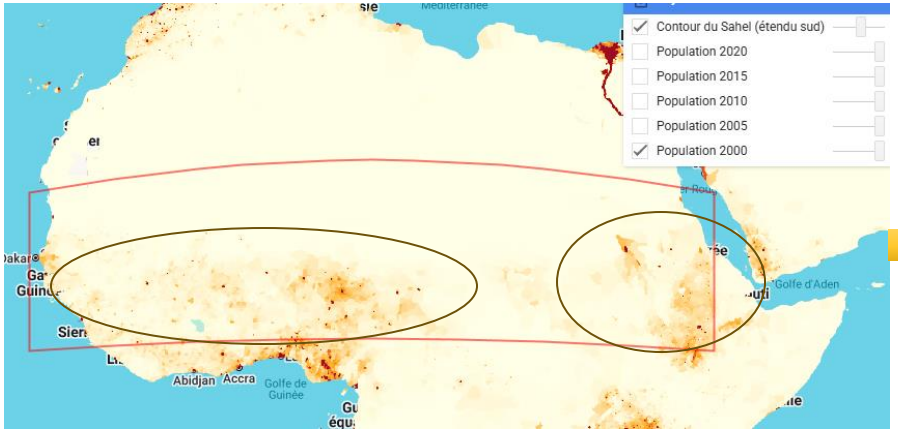
- Deepening the analysis of drought trends and vegetation loss.
- Investigating the impact on local populations & migration patterns.
- Exploring potential mitigation strategies using climate and hydrology data.

📚 **Lessons Learned**

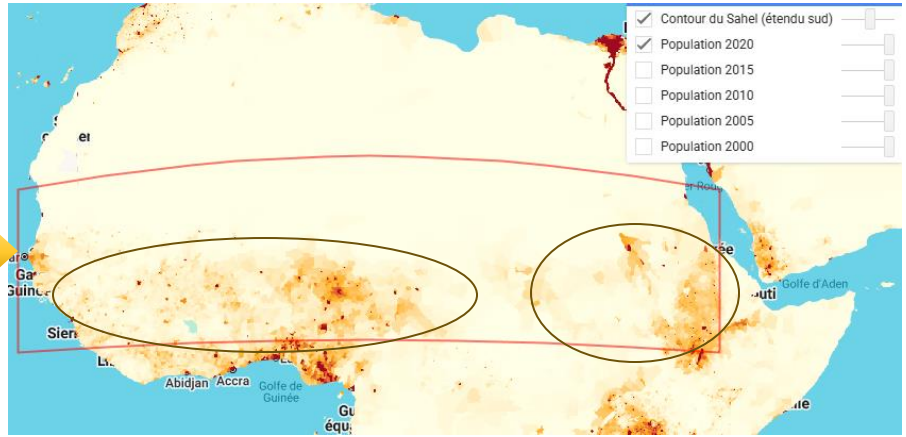
- ✅ Don’t rush research – Skipping steps can lead to mistakes that slow progress.
- ✅ Data interpretation challenges – Not all datasets are straightforward; careful analysis is crucial.
- ✅ Context matters – Understanding regional dynamics is essential for accurate conclusions.

Data Visualization – 1 – Population

2000



2020



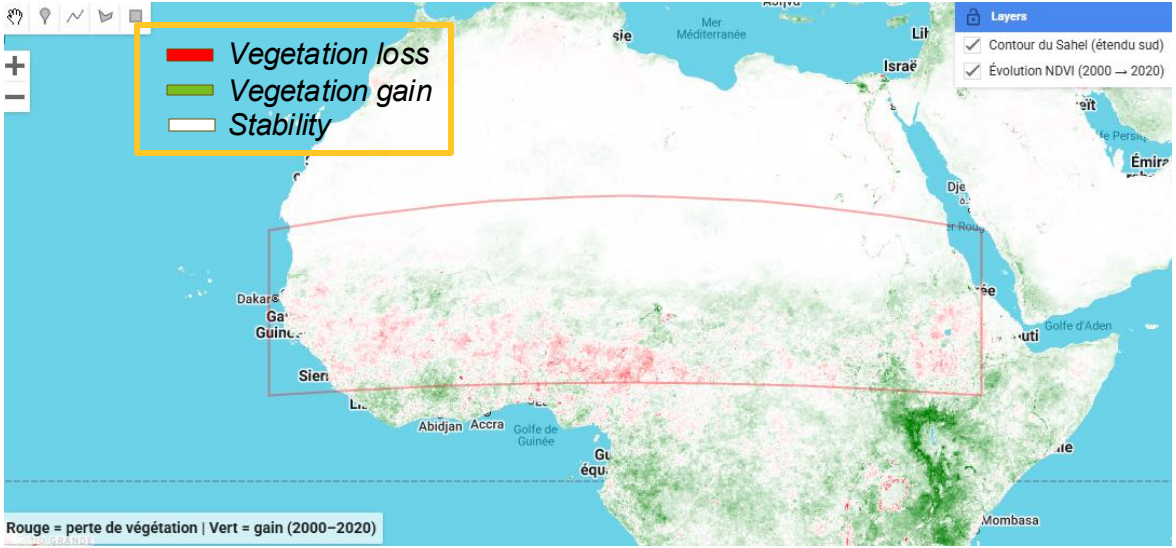
Map built using the : Center for International Earth Science Information Network (CIESIN) - (GPWv4.11)

Population Evolution from 2000 to 2020 in Sahelian Africa:
An increase in population has been observed in certain areas.
<u>Hypothesis</u>: These zones may offer a higher likelihood of water availability.

Population growth or migration ? That's the question

Data Visualization – 2 – Vegetation

Observing changes from 2000 to 2020 is challenging in the Sahel due to consistently **low NDVI*** values (0.1–0.3) resulting in minimal visual differences. Therefore, we built a difference map (2020–2000) to highlight areas of vegetation **loss**, **gain**, and **stability**.



NASA's DAAC LP at the USGS EROS Center

Questions raised from the observations : Is the change in vegetation correlated with population growth? Are there other factors we should consider before drawing any conclusions ?

Loss of vegetation ? Drought ? Does it explain the population evolution ?

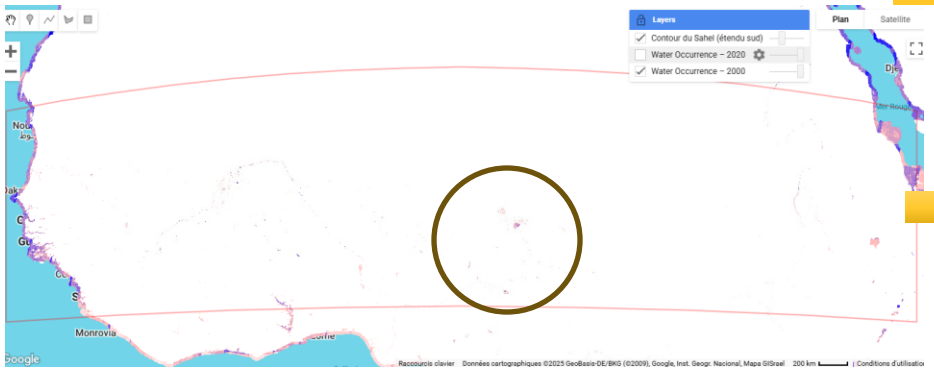
**NDVI (Normalized Difference Vegetation Index) is a satellite-derived metric that measures vegetation density and health.*

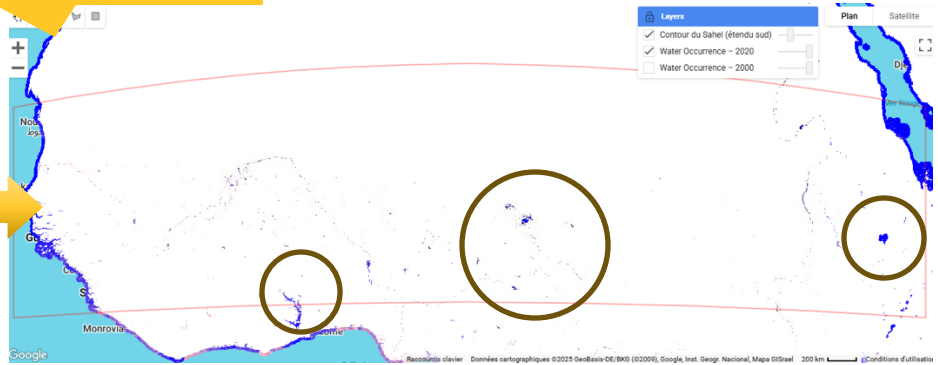
Data Visualization – 3 – Water

2000

The blue here corresponds to the rising sea levels along the coastline.

2020





Joint Research Centre (JRC) – Global Surface Water - Monthly History

Global surface water evolution between 2000 and 2020 :
We observe a surprising increase in the surface area of water. Notably, the Sahelian region of Africa, which is typically arid and prone to droughts, has shown significant changes. This raises an important question : could this be explained by a physical phenomenon ?

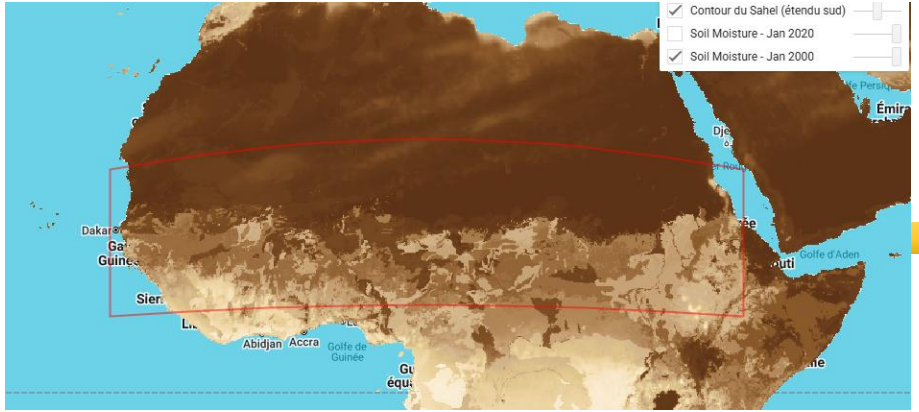
Surprising results : Increase in global surface water

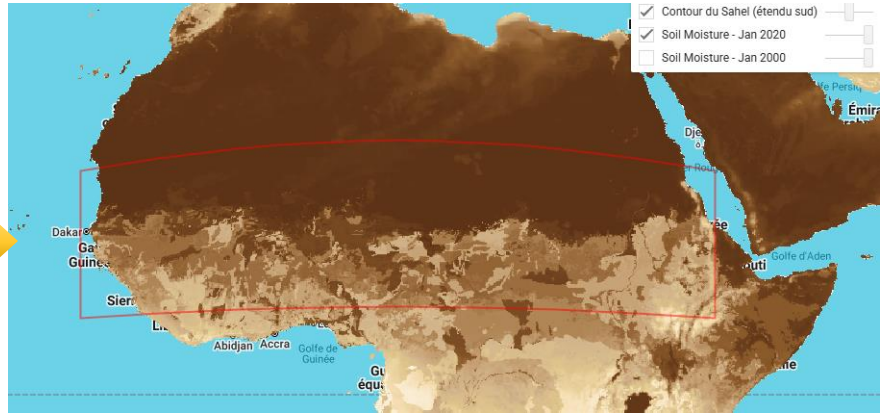
Data Visualization – 4 – Drought

2000

No Drought — Severe Drought

2020





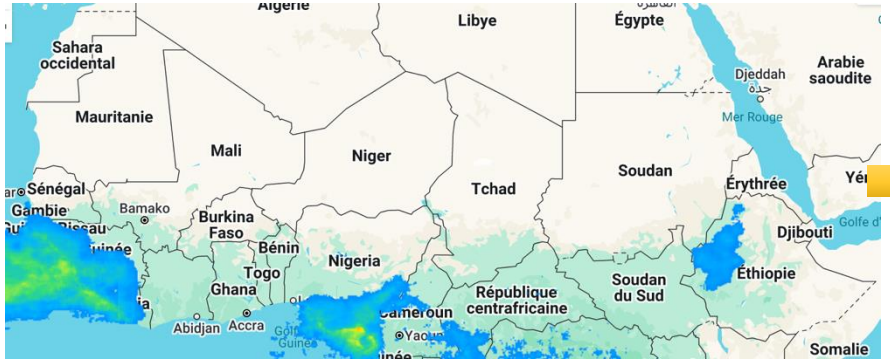
ERA5-Land Monthly Aggregated - ECMWF Climate

Drought evolution in the region between 2000 and 2020 :
As we know, this region has always been quite dry, but between 2000 and 2020, it has become even more arid, particularly from the north.
So, « why does the global surface water continue to rise ? » is the question we asked ourselves.

… But a growing drought ?

Data Visualization – 5 – Precipitation

2000



2020

Monthly Global Precipitation Measurement (GPM) v6

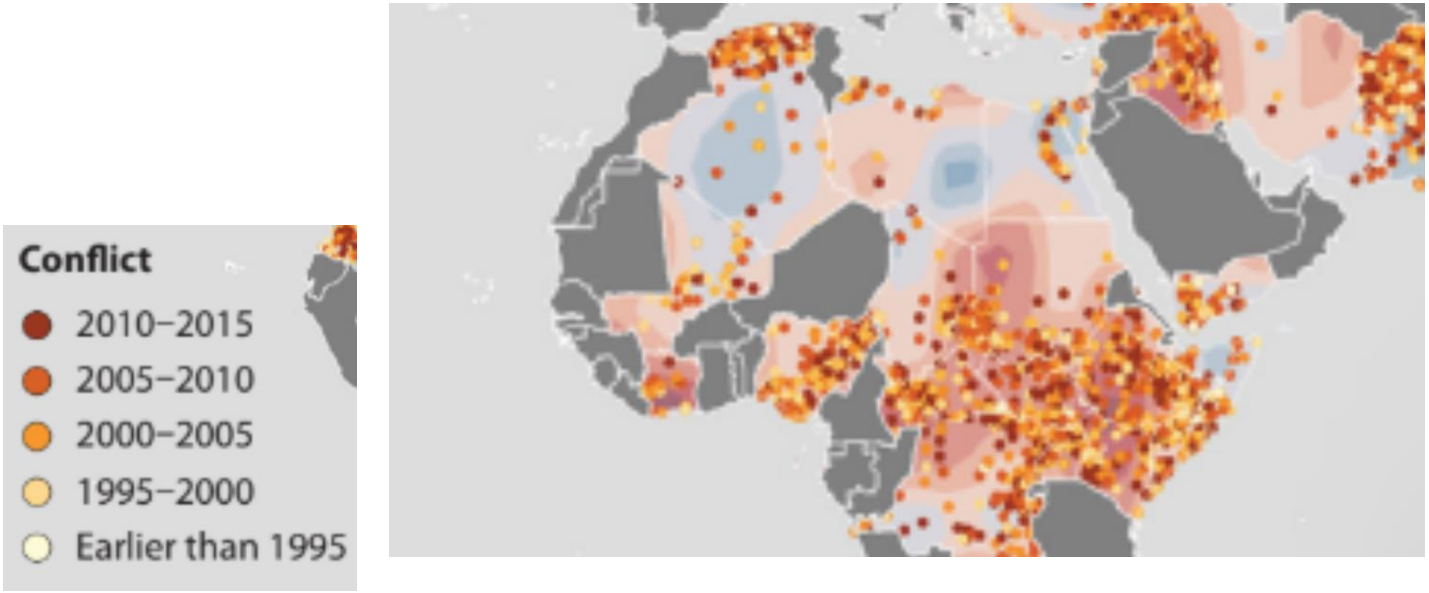
Precipitation evolution in the Region Between 2000 and 2020 :
As expected, precipitation has gradually decreased over time.
However, we've observed an increase in surface water. Let's explore why and what factors contribute to this rise.

A disappearing precipitation.

Analysis – 1

1.

Population escape from drought and search for water. But they also escape conflicts due to the lack of ressources, explaining the Evolution from 2000 to 2020



Analysis – 2

2.

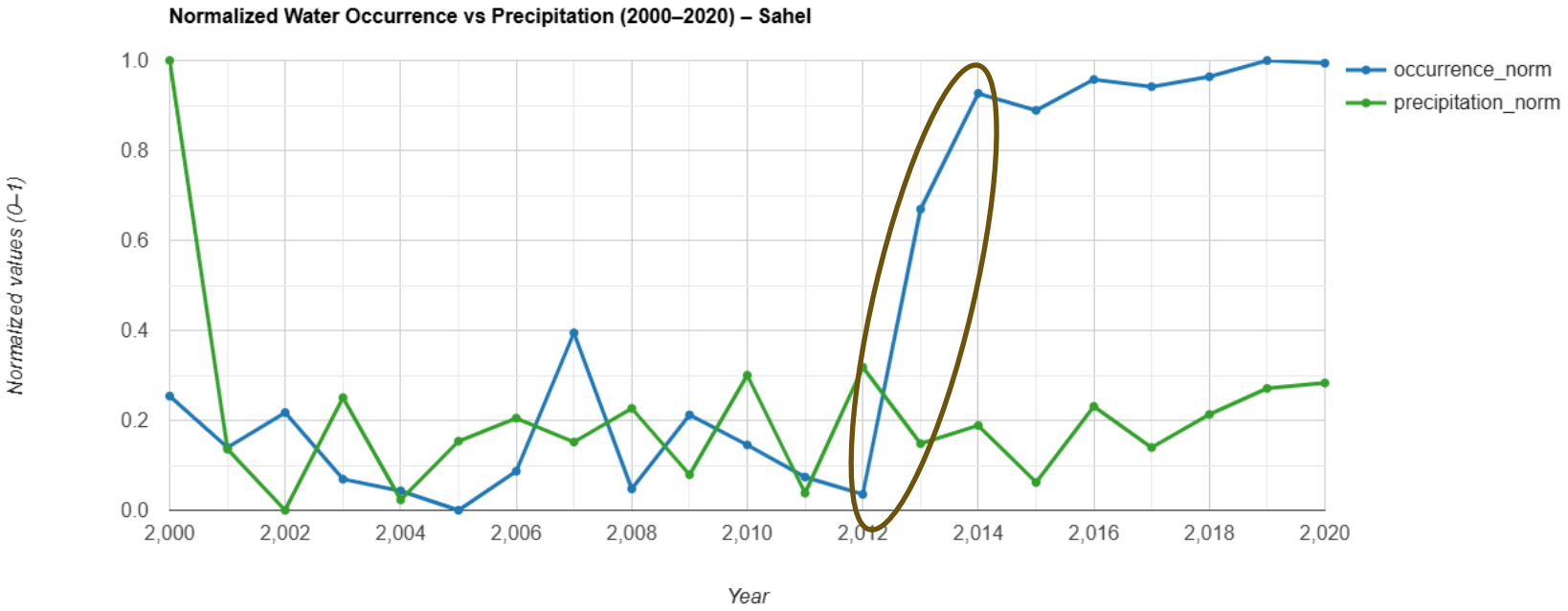
As the population increases or migrates, does vegetation drastically decrease ?

Our answer is : Yes. But we must consider several factors: population growth leads to **more buildings**, **less farmland**, and ultimately, **vegetation loss**. Simultaneously, the growing drought exacerbates the situation, further hindering vegetation growth. This cycle repeats itself, ultimately leading to conflicts over land and resource shortages caused by drought.

Quantitative Correlation – 1

3.

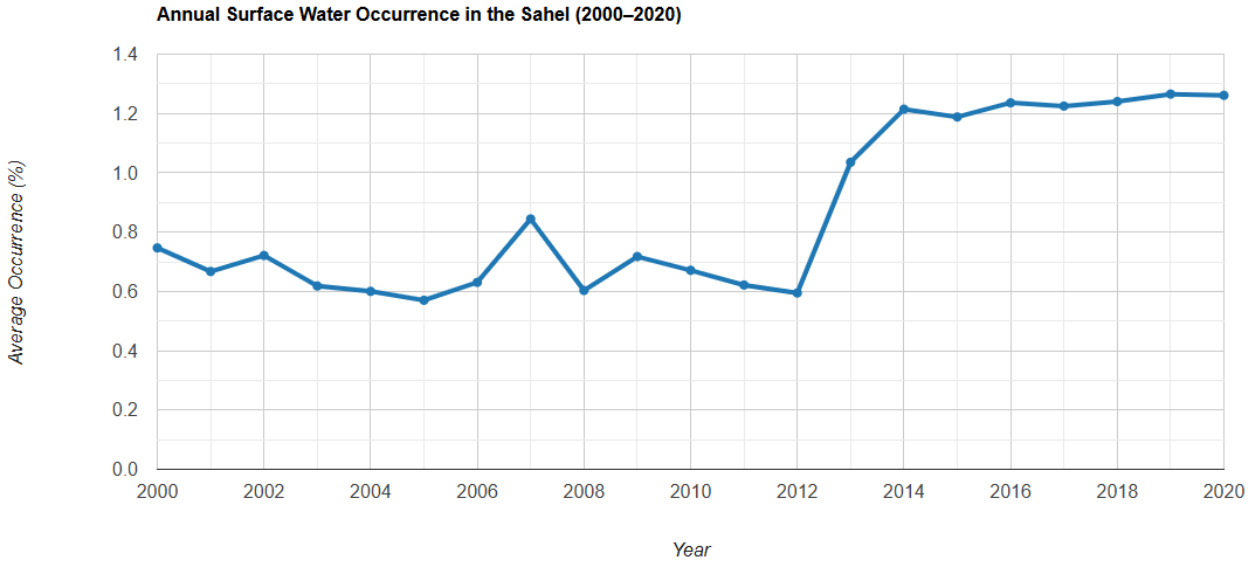
If we quantify our results regarding precipitation and global surface water, we notice several discrepancies. Typically, increased precipitation leads to more water in lakes and rivers for example. However, we don't observe a clear correlation between these values. On the contrary, since 2012, water levels have risen significantly, even though this seemed logical before.

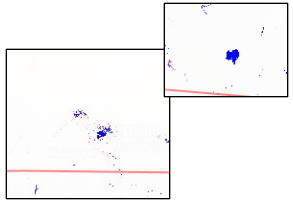


Analysis – 3 – Why ?

An increase in global surface water, despite the escalating aridity, can actually be explained by a simple factor often overlooked : the expansion of the sea along coastlines. According to *CNES*, this process continues to erode the land. Additionally, it pushed this even further between 2010 and 2012 : a significant precipitation impacted water levels in many regions, including Sahelian Africa (*AMMA-CATCH*).

Although we now understand this, we cannot claim that water evolution is solely influenced by the coastline. We can clearly observe the appearance of blue areas on the respective map. So why is this happening? It could be explained by a physical phenomenon.

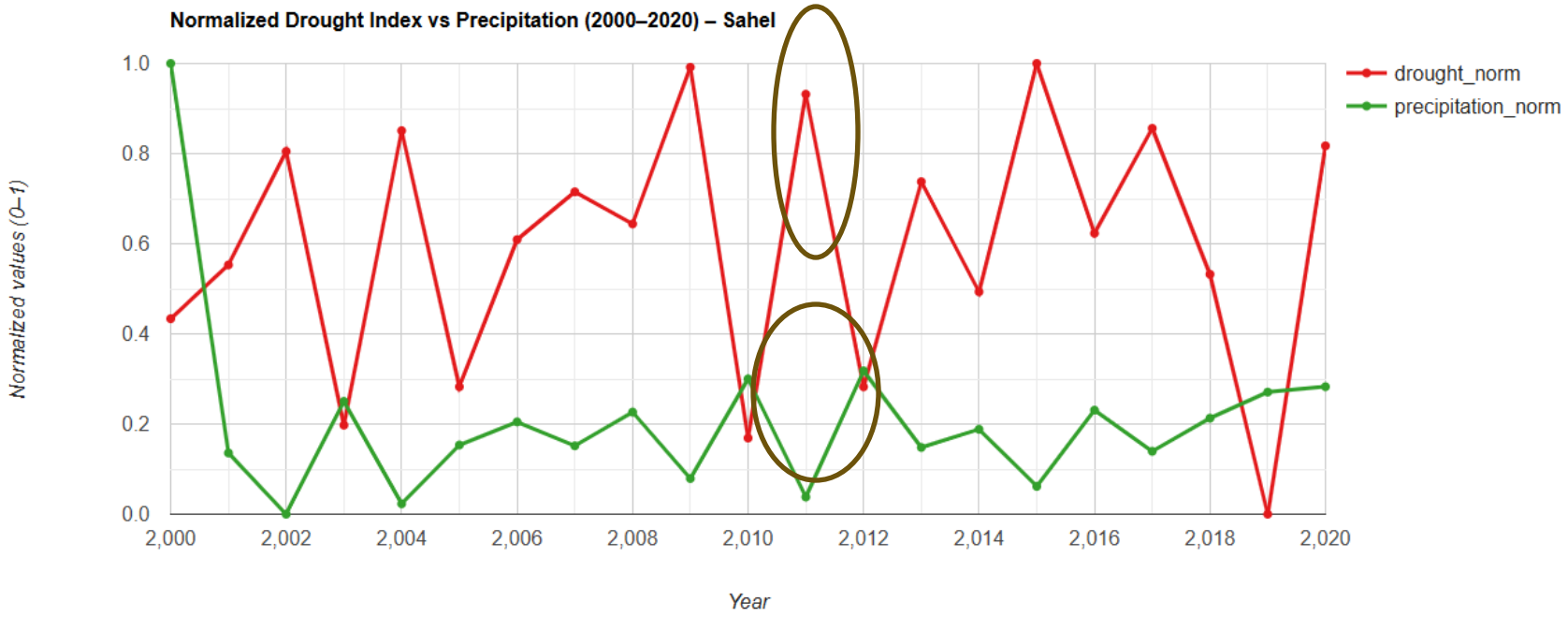




According to satellite-based datasets from JRC Global Surface Water, ERA5-Land (ECMWF), and CHIRPS (UCSB CHG), an increase in surface water extent does not necessarily indicate improved soil moisture or reduced drought conditions.

Analysis – 4

Conversely, as we can observe, drought and precipitation are generally correlated. When there's not that much of precipitation in a year, there's a bigger drought on the area, as we can observe in 2009.



Impact / Why

Our opinion is clear : satellite data can be incredibly useful in understanding and addressing the issue at hand, especially when it comes to **monitoring environmental changes** over time. However, it's essential to **integrate historical**, **meteorological**, and **local context** to avoid misinterpretation of the data. Satellite images alone can provide valuable insights, but without considering the bigger picture—such as historical climate patterns and human activity—there's a **risk** of making false correlations.

Using satellite data is powerful because it can offer a comprehensive, real-time overview of environmental changes. The key is ensuring we don't jump to conclusions based on partial data. By combining satellite data with **ground-level observations**, we can gain more accurate insights and avoid decisions based on misleading trends. It could lead to **better-informed policies**, especially in areas like **drought management**, **water conservation**, and **land use planning**.

Well, how can it impact policy changes ? Using satellite data could influence policy changes by providing undeniable, **large-scale evidence of environmental shifts**. For example, it could prompt authorities to **implement stricter water conservation measures** in areas showing **signs** of **increasing drought** or **decreasing vegetation**. Additionally, it can guide **urban planning**, as satellite data can highlight changes in land use or urban sprawl that affect **ecosystems**. Policymakers could use this data to create more proactive, **data-driven policies** for **sustainable** resource management.

Impact / Why

In Sahelian Africa, satellite data can reveal how drought and land degradation worsen **social issues** like displacement and poverty. By addressing these challenges with informed policies, we can better protect vulnerable populations and promote **environmental justice**.

Possible Implementation :

Satellite data could be used to monitor **land degradation** and **deforestation**, aligning with the UN SDG of life on land (SDG 15). By regularly tracking changes, we can better protect ecosystems and implement more effective reforestation efforts. Similarly, satellite monitoring could help track **water resources** (SDG 6), guiding efficient water use in drought-prone regions.

Further Study :

Ongoing research on how satellite data can track **seasonal** variations in precipitation, soil moisture, and vegetation health could help refine climate action strategies (SDG 13). Moreover, integrating this data into **climate models** could help forecast future changes, providing **early warnings** for governments and organizations to act on water scarcity or land loss. Studies could also explore how these satellite insights could be used to better design climate resilience strategies, particularly in **vulnerable** regions.

Thank you

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